GUEST LECTURE · CSIT321 · DATA WAREHOUSING & MINING

Data & AI: *The Honest Tour.*

What your textbook covers, what industry actually does,
and what you need to know before your first job interview.

60 minutes · no fluff · **bring questions**

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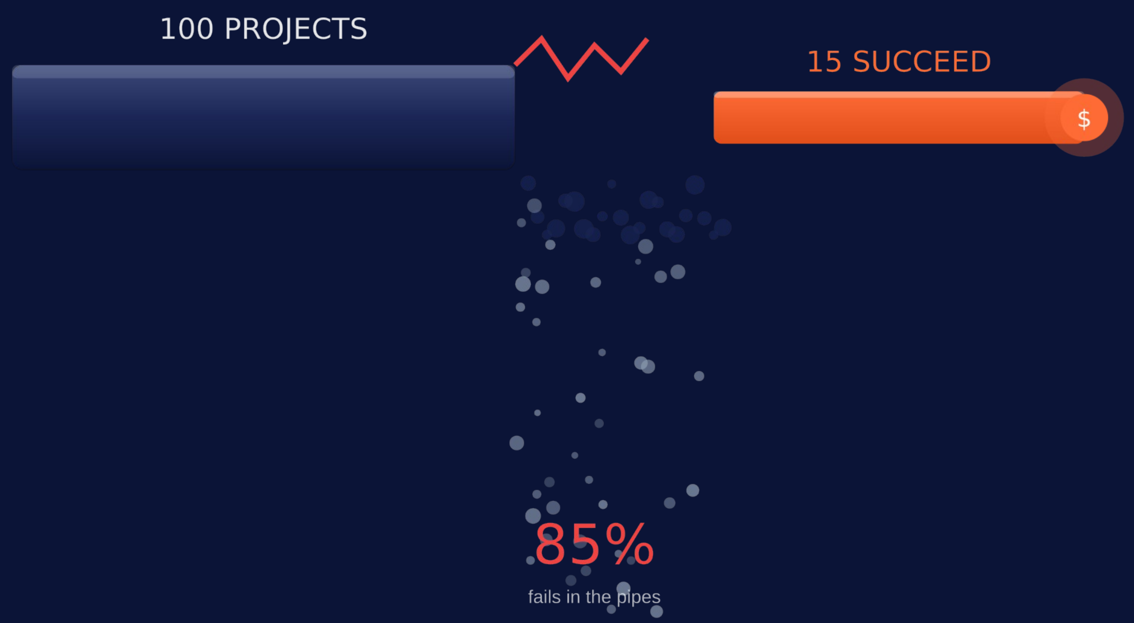
REALITY CHECK

85%

of data & AI projects fail to deliver business value.

Not because the tech is broken.
Because nobody asked 'why are we building this?'



The good news: the 15% that succeed are led by people who understand BOTH the pipes AND the business. *That's the job I'm describing today.*

THE NEXT 60 MINUTES

Here's the ride.

Six acts. Minimal textbook, maximum reality.

01 **The Landscape**
Why every company is now a data company

02 **Warehouse 101**
OLTP vs OLAP · the cube · why it still matters

03 **Dimensional Modeling**
Star · Snowflake · Galaxy · ROLAP/MOLAP/HOLAP

04 **Modern Stack**
ETL → ELT · Lake · Lakehouse · Medallion

05 **Data Mining → AI**
KDD · classification · clustering · road to GenAI

06 **Your Launchpad**
Free courses · student programs · certs that matter

01

THE LANDSCAPE

Why every company is now a data company.

In 2026, "non-tech company" is an extinct species.

THE LANDSCAPE

There is no such thing as a "non-tech company" in 2026.

Airline, bank, retailer, hospital — all quietly running a data/AI org the size of a small country.

$1.8T

global data & analytics spend by 2027

402 ZB

of data will be created in 2026

2.5M+

open data/AI roles worldwide

90%

of the world's data is from post-2022

Translation: *the market isn't short on demand. It's short on people who know what they're doing.*

WHERE CS GRADS LAND

Six roles, one data ecosystem.

You'll touch at least three of these in your first 3 years.

Data Engineer

"the plumber"

Builds the pipes. Moves data. Fixes it at 2am.

₹8–25 L · $25–75K

Analytics Engineer

"the translator"

Turns raw data into trusted business metrics.

₹10–28 L · $30–90K

Data Analyst / BI

"the storyteller"

Talks to the business. Builds dashboards. Asks 'why'.

₹6–18 L · $20–60K

Data Scientist

"the detective"

Statistical models, A/B tests, forecasting.

₹12–35 L · $40–110K

ML / AI Engineer

"the operator"

Ships models to prod. Tunes LLMs. Fights drift.

₹15–50 L · $60–180K

Data Architect

"the CTO's CTO"

Designs the whole house. The one the CTO calls.

₹25–70 L · $100–250K

Early-career ranges · India (INR lakhs/yr) and global (USD/yr). Blend of Glassdoor, LinkedIn, community data.

02

WAREHOUSE 101

Why OLTP can't do what OLAP does.

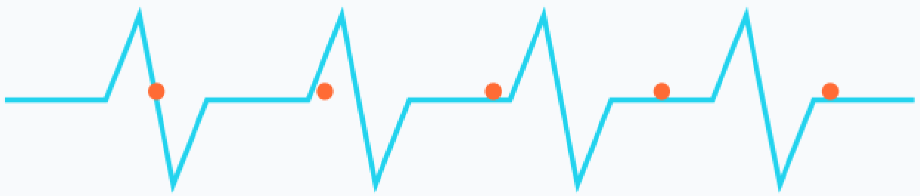
Module I · syllabus deep-dive starts here.

OLTP vs OLAP — same SQL, completely different brains.

One runs the business. One understands it. Mix them up → your bank crashes at 11 a.m.

OLTP

RUN THE BUSINESS · milliseconds · many tiny writes

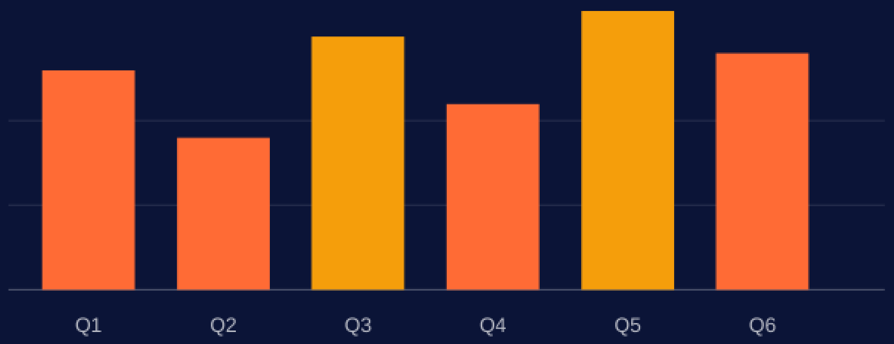


```
INSERT order #9371
UPDATE stock -1
INSERT payment
UPDATE loyalty pts
INSERT shipping
```

"the panicked waiter taking 400 orders per minute"

OLAP

UNDERSTAND THE BUSINESS · seconds · massive reads




```
SELECT region, SUM(amount) FROM fact_sales GROUP BY ...
```

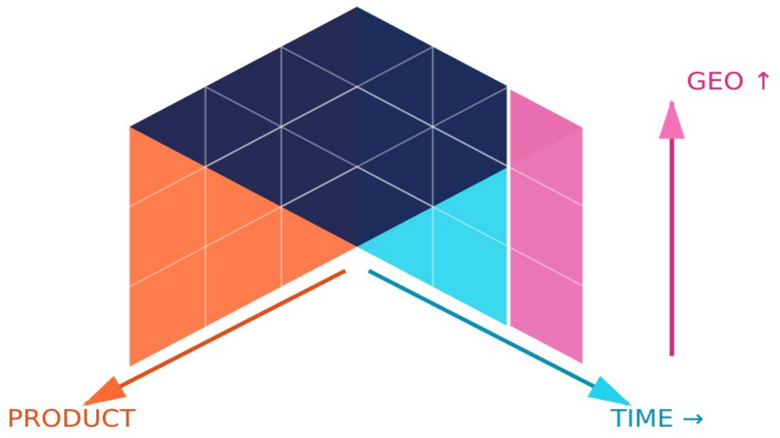
"the chef studying last quarter's receipts to rewrite the menu"

WAR STORY · *the intern ran SELECT SUM(amount) FROM transactions on prod and crashed the bank at 11 a.m. Every senior has lived this.*

MULTIDIMENSIONAL THINKING

The Data Cube — reality is 3D+.

Rows & columns are 2D. Business questions aren't. That's the whole idea.



THE 5 MOVES ON A CUBE

SLICE
Fix one dim → 'Sales in Q3'

DICE
Fix many dims → 'Q3, India, Mobile'

DRILL-DOWN
Go deeper → Year → Month → Day

ROLL-UP
Go higher → Day → Month → Year

PIVOT
Rotate the cube → swap axes

MODERN ANGLE ·
You won't build a literal cube in 2026. But every PivotTable, every Looker dashboard, every 'Slice by region' button is the cube in disguise.

03

DIMENSIONAL MODELING

How to shape data so humans can ask it questions.

Module II · the part that makes or breaks a warehouse.

DIMENSIONAL MODELING

Facts vs Dimensions — the mental model.

A fact is something that happened. A dimension is the context around it.

> **Ravi** bought **2 iPhones** on **Tuesday** at the **Dubai Mall** for **AED 7,400**.

- CUSTOMER (dim)
- QTY (fact)
- DATE (dim)
- STORE (dim)
- AMOUNT (fact)

FACT_SALES

```
sale_id (PK)
customer_id (FK)
product_id (FK)
date_id (FK)
store_id (FK)

quantity → MEASURE
amount → MEASURE
```

DIM_CUSTOMER

```
customer_id (PK)
name
age
tier
city
country
```

DIM_PRODUCT

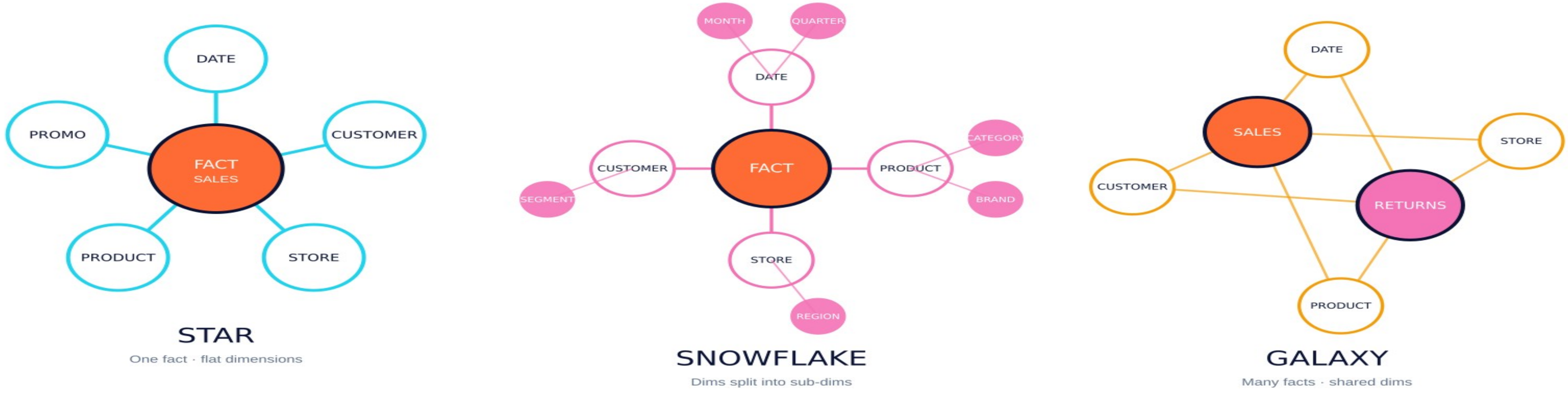
```
product_id (PK)
name
category
brand
price
sku
```

DIM_DATE

```
date_id (PK)
day
day_of_week
month
quarter
year
is_weekend
is_holiday
```

Star · Snowflake · Galaxy.

Three ways to arrange facts-and-dims. Trade-offs: speed vs storage vs complexity.



INDUSTRY RULE OF THUMB · **start STAR.** Snowflake *only* if a dim is huge and rarely changes. Galaxy arrives on its own when the business grows.

OLAP STORAGE MODELS

ROLAP · MOLAP · HOLAP.

Your textbook treats them as equals. Industry has picked a winner.

ROLAP

★ WON

Relational OLAP

STORAGE

Relational DB (tables)

✓ PRO

Scales to huge data

✗ CON

Slower queries

TODAY

Snowflake, BigQuery, Redshift, Databricks — all ROLAP descendants

MOLAP

Multidimensional OLAP

STORAGE

Proprietary cube files

✓ PRO

Blazing-fast reads

✗ CON

Rebuilds, storage cost

TODAY

SSAS cubes, Essbase — mostly legacy

HOLAP

Hybrid OLAP

STORAGE

Cube for hot + DB for cold

✓ PRO

Best of both

✗ CON

Complex to run

TODAY

Mostly replaced by cloud warehouses with caching

REALITY · *cloud warehouses use ROLAP storage + columnar/in-memory tricks to feel like MOLAP. You get both. You pay per query. You go home.*

04

THE MODERN STACK

From ETL to Lakehouse in one evolution story.

The part that's NOT in your textbook but IS in every job description.

THE MODERN STACK

ETL vs ELT — the order of operations changed everything.

Same three letters. Different order. Billion-dollar consequence.

ETL · the old way (pre-2015)

SOURCES CRM, ERP, files → **EXTRACT** → **TRANSFORM** on a lonely server → **LOAD** into warehouse → **WAREHOUSE** rigid · slow

ELT · the modern way (post-2015) — *after cloud compute got cheap*

SOURCES anything, any format → **EXTRACT** → **LOAD** as-is, raw → **LAKEHOUSE** → **TRANSFORM** inside, with SQL

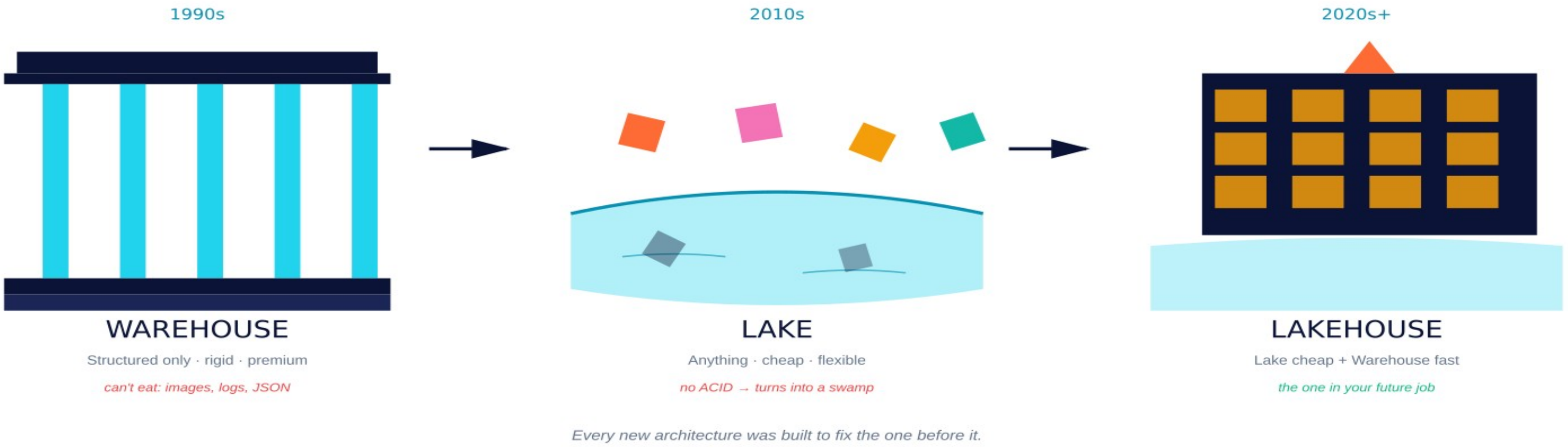
ETL · transform before storing. Rigid, slow, but disciplined.

ELT · load everything, transform on demand. Flexible, fast, cheap.

20 YEARS IN ONE SLIDE

Warehouse → Lake → Lakehouse.

Each new architecture was built to fix the last one's problem.



TODAY'S LAKEHOUSE PLAYERS

- Databricks + Delta
- Snowflake + Iceberg
- Azure Fabric / OneLake
- BigQuery + BigLake
- AWS S3 + Iceberg / Hudi

THE HERO SLIDE

Medallion Architecture.

Bronze → Silver → Gold. Three layers, one pipeline, ruthless discipline.

↑ If you remember ONE diagram from today, let it be this one. ↑

If you remember ONE diagram from this lecture, let it be this one. *Every modern platform — Databricks, Snowflake, Fabric, BigQuery — is a dialect of Bronze→Silver→Gold.*

MODERN DATA STACK · FIELD MAP

You don't need to know every tool. You need to know which tool solves which problem.

INGESTION

move bytes in

- Fivetran
- Airbyte
- Kafka
- Debezium

STORAGE

hold them cheap

- S3 / ADLS / GCS
- Delta
- Iceberg
- Hudi

COMPUTE / SQL

query fast

- Databricks
- Snowflake
- BigQuery
- DuckDB

TRANSFORM

shape the data

- dbt
- SQLMesh
- Spark
- Dataform

ORCHESTRATION

run jobs on time

- Airflow
- Dagster
- Prefect
- Mage

BI / ANALYTICS

humans see it

- Power BI
- Tableau
- Looker
- Metabase

ML / AI

train + serve

- MLflow
- Vertex AI
- SageMaker
- Ray

OBSERVABILITY

catch bad data

- Monte Carlo
- Great Expectations
- Soda
- Elementary

CAREER TIP · master ONE tool per category and you're employable anywhere. Nobody uses all 32 — every team picks 6–8.

05

DATA MINING → AI

KDD, classification, clustering, and the road to GenAI.

Modules III, IV, V — the punchline of your syllabus.

THE KDD PROCESS

Knowledge Discovery in Databases.

Data mining is step 4 of 6. The other 5 are where the work actually lives.

WHERE THE TIME ACTUALLY GOES

70%

unglamorous work.
Cleaning, joining, coaxing.

The other 30%?

Modeling (15%) + Shipping (15%)

The '70% data prep' stat
is industry folklore, and roughly true. Accept it early — it makes you happier.

MODULE IV IN ONE SLIDE

The 5 techniques · one cheat sheet.

When a business problem lands on your desk, you'll pattern-match to one of these.

CLASSIFICATION

What class is this?

Is this spam? Will this loan default?

`Logistic · Tree · RF · XGBoost · NN`

REGRESSION

How much? How many?

What will revenue be next month?

`Linear · RF · GBM · Prophet`

CLUSTERING

Who belongs together?

Segment 40M customers into 8 personas

`K-means · DBSCAN · Hierarchical`

ASSOCIATION

What goes with what?

Beer + diapers. Bread + butter.

`Apriori · FP-Growth`

ANOMALY

What's weird here?

Fraud detection. Server outages.

`Isolation Forest · Autoencoder`

ORIGIN STORY · BEER + DIAPERS

1990s: someone ran association rules on Walmart receipts. Men who bought diapers also bought beer. That's Market Basket Analysis. That's also how Netflix recommends your next show.

30 YEARS, ONE FIELD

Data Mining → ML → Deep Learning → GenAI → Agentic AI.

If you understand mining, ML makes sense. If ML makes sense, GenAI isn't magic.

THE PLOT TWIST

GenAI didn't replace the warehouse — it made it matter MORE.

LLMs are useless without your clean, governed Gold layer to ground them. RAG without clean data is a confident liar.

MODULE V IN REALITY

Where this ALL actually ships.

Real industries. Real dollars. Problems being solved right now.

RETAIL / E-COM

basket · recs · demand forecasting

Amazon: 35% of revenue from recommendation engine

BANKING

fraud · credit · AML

HDFC blocks millions of fraudulent txns/year with ML

TELECOM

churn · network optimization

Jio predicts churn 90 days out with ML

HEALTHCARE

diagnosis · drug discovery · imaging

AI reads chest X-rays faster than radiologists

LOGISTICS

routing · demand · warehousing

Swiggy/Zomato ETAs are pure ML

MEDIA / SOCIAL

recs · moderation · ranking

Netflix, YouTube, Instagram = recommendation engines

THINGS YOUR TEXTBOOK WILL NOT TELL YOU

War stories · how this breaks in production.

Three failure modes. Nobody teaches them. Everybody hits them.

SCHEMA DRIFT

THE SCENE

Upstream team renames a column. Your pipeline silently writes NULL for 3 weeks. CFO notices in a board meeting.

THE LESSON

Validate schemas at every boundary. Great Expectations · Soda · dbt tests.

SILENT QUALITY

THE SCENE

Source feed doubles a record. Dashboard shows 2× revenue. Everyone high-fives. Until the auditor shows up.

THE LESSON

Monitor row counts, freshness, and distributions. If it moves 3σ, alert.

MODEL DECAY

THE SCENE

Fraud model was 94% in training. Six months later it's 61% and nobody noticed — the dashboard was static.

THE LESSON

Every model needs a drift monitor + retrain cadence from day one.

COMMON THREAD · *data quality is not a stage. It's a religion. Build monitoring BEFORE you build dashboards.*

06

YOUR LAUNCHPAD

Free courses, student programs, and the certs that matter.

Screenshot this section.

FREE COURSES

Free courses from the OEMs themselves.

These companies WANT you certified. They give it away. Use them.

DATABRICKS

Databricks Academy

Lakehouse, Spark, Delta, MLflow — all free

`customer-academy.databricks.com`

SNOWFLAKE

Snowflake University

Zero-to-hero SQL + warehouse fundamentals

`learn.snowflake.com`

MICROSOFT

Learn · Fabric / Azure

DP-203, DP-600, AI-900 prep free

`learn.microsoft.com`

GOOGLE CLOUD

Cloud Skills Boost

BigQuery, Dataproc, Vertex AI labs

`cloudskillsboost.google`

AWS

AWS Skill Builder

Redshift, Glue, SageMaker fundamentals

`skillbuilder.aws`

NVIDIA

Deep Learning Institute

GPU computing, LLMs, RAG, Triton

`nvidia.com/dli`

HUGGING FACE

HF Learn

LLMs, fine-tuning, agents, RL — gold tier

`huggingface.co/learn`

DEEPLEARNING.AI

Andrew Ng courses

ML, deep learning, GenAI, MLOps

`deeplearning.ai`

STUDENT PROGRAMS

Your .edu email unlocks thousands in cloud credits.

It expires when you graduate. Claim it THIS WEEK.

GitHub Student Developer Pack

Copilot Pro, JetBrains, 100+ premium tools

$200K+

AWS Educate

Free training + AWS cloud credits

$50–$200

Google Cloud for Students

Skills Boost subscription + credits on request

$300 + extras

Microsoft Azure for Students

$100 credit, no credit card, free dev tools

$100 renewable

Databricks Community Edition

Free Spark + notebook cluster

Unlimited · free forever

Snowflake Student Trial

120 days of Snowflake + training path

$400 credits · 120d

NVIDIA DLI Teaching Kits

Self-paced DL/GenAI courses + certs

Free, certified

Kaggle

Also: your portfolio. Competitions + datasets

Free GPUs/TPUs

DO THIS THIS WEEK · *every semester you delay, you're leaving thousands of dollars of compute on the table.*

CERTS + COMMUNITIES

The certs that matter · the rooms that keep you current.

Not an exhaustive list. A deliberately short one.

CERTS THAT MATTER IN 2026

- Databricks — Data Engineer Associate
- Azure — DP-700 Fabric Analytics Engineer
- AWS — Data Engineer Associate (DEA-C01)
- Google Cloud — Professional Data Engineer
- Snowflake — SnowPro Core
- NVIDIA — Generative AI with LLMs
- dbt — dbt Analytics Engineering Cert

COMMUNITIES TO STAY CURRENT

- r/dataengineering · r/MachineLearning
- Locally Optimistic (Slack) — analytics eng
- dbt Slack — dbt + modern data stack
- MLOps Community (Slack)
- Hugging Face Discord
- Kaggle forums
- Podcasts: Data Engineering · Latent Space · DataFramed

TRUTH BOMB · *build ONE end-to-end project, push to GitHub, write it up on LinkedIn — stands out more than any cert.*

5 portfolio projects that don't suck.

Each teaches one concept from this lecture. Each looks great on a resume.

1	**Mini Medallion Warehouse** `Python + DuckDB + dbt + Metabase`	TEACHES **Bronze/Silver/Gold, ELT, BI**
2	**Streaming Fraud Detector** `Kafka + Spark + Delta + Streamlit`	TEACHES **Real-time pipes, anomaly detection**
3	**Retail Recommender** `Apriori / ALS on MovieLens`	TEACHES **Association rules, collaborative filtering**
4	**Customer Churn Predictor** `pandas + sklearn + SHAP + FastAPI`	TEACHES **Classification, feature eng, serving**
5	**RAG over your college PDFs** `LangChain + Chroma + Llama / GPT`	TEACHES **Embeddings, RAG, GenAI basics**

Don't try to build a unicorn. *Ship something small, end-to-end, with a README and a Loom video. That IS a portfolio.*

IF YOU REMEMBER TEN THINGS

10 things I want you to walk out remembering.

01 OLTP runs the business. OLAP understands it. Don't mix them.

02 Start with a STAR schema. Everyone does. Everyone should.

03 ELT replaced ETL because cloud compute became cheaper than transform servers.

04 Bronze → Silver → Gold is the default mental model. Memorize it.

05 70% of a data scientist's job is cleaning data. Accept it early.

06 Classification · Regression · Clustering · Association · Anomaly — match the question to the tool.

07 GenAI made data quality MORE important, not less. RAG without clean data is a liar.

08 Data quality is a religion, not a stage. Monitor schemas and distributions always.

09 Pick ONE cloud stack and go deep. T-shaped > shallow across 10.

10 Build ONE end-to-end project. Ship it. The README matters more than the stack.

ONE MORE THING.

The tools will change.

The patterns won't.

In five years, half the tool names on the earlier slides will be gone — replaced by better ones. But OLTP vs OLAP, facts vs dimensions, batch vs streaming, Bronze vs Gold — those are forever. Learn the patterns. The tools are just the accent.

Thank you. *Next session — we build a mini Medallion warehouse from scratch. SQLite, Python, React. Bring your laptop.*

Q&A

ask the dumb questions. that's the whole point.